

Physics 7501 Notes

Sharon Ye

1 Linear Algebra Review

1.1 Vector Spaces and Hilbert Spaces

Definition 1.1. A *vector space* over a field $\mathbb{F}$ is a set V equipped with two binary operations: addition and scalar multiplication, and for which the following properties hold

- (1) Closure under addition: $|u\rangle + |v\rangle \in V$, $\forall |u\rangle, |v\rangle \in V$
- (2) Closure under scalar multiplication: $\alpha |v\rangle \in V$, $\forall \alpha \in \mathbb{F}, |v\rangle \in V$
- (3) Additive identity: $\exists |0\rangle \in V$ such that $|v\rangle + |0\rangle = |v\rangle$, $\forall |v\rangle \in V$
- (4) Additive inverses: $\exists |u\rangle \in V$ such that $|v\rangle + |u\rangle = |0\rangle$, $\forall |v\rangle \in V$
- (5) Multiplicative scalar identity: $\exists 1 \in \mathbb{F}$ such that $1 |v\rangle = |v\rangle$, $\forall |v\rangle \in V$
- (6) Associativity of addition: $(|u\rangle + |v\rangle) + |w\rangle = |u\rangle + (|v\rangle + |w\rangle)$, $\forall |u\rangle, |v\rangle, |w\rangle \in V$
- (7) Commutativity of addition: $|u\rangle + |v\rangle = |v\rangle + |u\rangle$, $\forall |u\rangle, |v\rangle \in V$
- (8) Distributivity of over vector addition: $\alpha(|u\rangle + |v\rangle) = \alpha |u\rangle + \alpha |v\rangle$, $\forall \alpha \in \mathbb{F}$ and $|u\rangle, |v\rangle \in V$
- (9) Distributivity over scalar addition: $(\alpha + \beta) |v\rangle = \alpha |v\rangle + \beta |v\rangle$, $\forall \alpha, \beta \in \mathbb{F}, |v\rangle \in V$
- (10) Compatibility of scalar multiplication: $\alpha(\beta |v\rangle) = (\alpha\beta) |v\rangle$, $\forall \alpha, \beta \in \mathbb{F}, |v\rangle \in V$

Definition 1.2. A *Hilbert space* is a vector space over $\mathbb{C}$ that is equipped with an inner product $\langle u|v\rangle \in \mathbb{C}$ with the following properties

- (1) Conjugate symmetry: $\langle u|v\rangle = \langle v|u\rangle^*$
- (2) Linearity in the ket: $\langle u|(a|v\rangle + b|w\rangle) = a\langle u|v\rangle + b\langle u|w\rangle$
- (3) Positive definiteness: $\langle v|v\rangle \geq 0$, $\langle v|v\rangle = 0 \iff |v\rangle = |0\rangle$

and is also complete (every Cauchy sequence converges).

Examples of Hilbert spaces:

- (i) n -dimensional Hilbert space on complex numbers: $V \simeq \mathbb{C}^n$
- (ii) $n \times n$ complex matrices equipped with the Hilbert-Schmidt inner product forms a Hilbert space $L_{\mathbb{C}^n}$ (see Homework 1)
- (iii) n -dimensional Hilbert space on real numbers: $V \simeq \mathbb{R}^n$

Examples of Non-Hilbert spaces:

- (i) n -dimensional vector space $V \simeq \mathbb{Q}^n$ on rational numbers is not a Hilbert space because it is not complete, since a Cauchy sequence of rational numbers can converge to an irrational number.
- (ii) The inner product space of continuous square-integrable complex functions on interval $[a, b]$ with the following L^2 inner product:

$$\langle f_1 | f_2 \rangle \equiv \int_a^b f_1^*(x) f_2(x) dx \quad (1.39)$$

is not a Hilbert space due to its incompleteness: a Cauchy sequence of continuous functions can be discontinuous. Its completion is the Hilbert space $L^2[a, b]$.

- (iii) For the same reason of incompleteness, the inner product space of polynomials on $[a, b]$ with the same L^2 inner product is not a Hilbert space, even though it's a subspace of $L^2[a, b]$.

1.2 Cauchy-Schwartz and Triangle Inequality

Theorem 1.1. Cauchy-Schwartz Inequality: $\langle v | v \rangle \langle u | u \rangle \geq |\langle u | v \rangle|^2$

Proof. If $|u\rangle = |0\rangle$, both sides are zero. Otherwise, define

$$|w\rangle = |v\rangle - \lambda |u\rangle, \quad \lambda = \frac{\langle u | v \rangle}{\langle u | u \rangle}.$$

By positivity of the inner product,

$$\begin{aligned} 0 \leq \langle w | w \rangle &= (\langle v | - \lambda^* \langle u |) (|v\rangle - \lambda |u\rangle) \\ &= \langle v | v \rangle - \lambda \langle v | u \rangle - \lambda^* \langle u | v \rangle + |\lambda|^2 \langle u | u \rangle. \end{aligned}$$

Using $\langle v | u \rangle = \langle u | v \rangle^*$ and substituting λ ,

$$\begin{aligned} 0 \leq \langle v | v \rangle - \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle} - \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle} + \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle} \\ = \langle v | v \rangle - \frac{|\langle u | v \rangle|^2}{\langle u | u \rangle}. \end{aligned}$$

Multiplying by $\langle u | u \rangle > 0$ gives the inequality. Equality holds precisely when $\langle w | w \rangle = 0$, which, by positive definiteness, means

$$|w\rangle = |0\rangle \iff |v\rangle = \lambda |u\rangle.$$

Theorem 1.2. Triangle Inequality: $\| |v\rangle + |w\rangle \| \leq \| |v\rangle \| + \| |w\rangle \|$

Proof. Expand the square of the norm:

$$\begin{aligned}\| |v\rangle + |w\rangle \|^2 &= (\langle v| + \langle w|)(|v\rangle + |w\rangle) \\ &= \langle v|v\rangle + \langle w|w\rangle + \langle v|w\rangle + \langle w|v\rangle \\ &= \| |v\rangle \|^2 + \| |w\rangle \|^2 + 2 \operatorname{Re}\langle v|w\rangle.\end{aligned}$$

Since $\operatorname{Re}(z) \leq |z|$, and Cauchy–Schwarz gives

$$|\langle v|w\rangle| \leq \sqrt{\langle v|v\rangle\langle w|w\rangle} = \| |v\rangle \| \| |w\rangle \|,$$

we obtain

$$\begin{aligned}\| |v\rangle + |w\rangle \|^2 &\leq \| |v\rangle \|^2 + \| |w\rangle \|^2 + 2\| |v\rangle \| \| |w\rangle \| \\ &= (\| |v\rangle \| + \| |w\rangle \|)^2.\end{aligned}$$

Taking the non-negative square root proves the triangle inequality.

1.3 Hilbert-Schmidt Inner Product

Definition 1.3. The *Hilbert-Schmidt Inner Product* is a binary operation on $\mathbb{L}_V \times \mathbb{L}_V \rightarrow \mathbb{C}$ defined by

$$(A, B) \equiv \operatorname{tr}(A^\dagger B) \quad A, B \in \mathbb{L}_V$$

Proof. To show that this is indeed an inner product, we need the following properties:

(1) Positive definiteness:

$$(A, A) = \operatorname{tr}(A^\dagger A) = \sum_{i=1}^n \sum_{j=1}^n (A^\dagger)_{ij} A_{ji} = \sum_{i=1}^n \sum_{j=1}^n |A_{ji}|^2 \geq 0,$$

and

$$\sum_{i=1}^n \sum_{j=1}^n |A_{ji}|^2 = 0 \iff A_{ji} = 0 \forall i, j \implies A = 0.$$

(2) Linearity in the second argument:

$$\begin{aligned}(A, \alpha B + \beta C) &= \operatorname{tr}(A^\dagger(\alpha B + \beta C)) \\ &= \operatorname{tr}(\alpha A^\dagger B + \beta A^\dagger C) \\ &= \alpha \operatorname{tr}(A^\dagger B) + \beta \operatorname{tr}(A^\dagger C) \\ &= \alpha(A, B) + \beta(A, C).\end{aligned}$$

(3) Conjugate linearity in first argument / conjugate symmetry:

$$\begin{aligned}(\alpha A + \beta B, C) &= \operatorname{tr}((\alpha A + \beta B)^\dagger C) \\ &= \operatorname{tr}(\alpha^* A^\dagger C + \beta^* B^\dagger C) \\ &= \alpha^*(A, C) + \beta^*(B, C).\end{aligned}$$

$$\begin{aligned}(A, B) &= \operatorname{tr}(A^\dagger B) = \operatorname{tr}((B^\dagger A)^\dagger) \\ &= (\operatorname{tr}(B^\dagger A))^* = (B, A)^*.\end{aligned}$$

Under the Hilbert-Schmidt inner product, the 3 Pauli matrices $\sigma_x, \sigma_y, \sigma_z$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and the 2×2 identity matrix $\hat{1}$ together can form a complete orthonormal basis for the vector space $L_{\mathbb{C}^2}$ of all 2×2 complex matrices.

Proof. We know that

$$\sigma_x \sigma_y = i \sigma_z, \quad \sigma_y \sigma_z = i \sigma_x, \quad \sigma_z \sigma_x = i \sigma_y.$$

$\implies$

$$\text{tr}(\sigma_x \sigma_y) = i \text{tr}(\sigma_z) = 0,$$

$$\text{tr}(\sigma_y \sigma_z) = i \text{tr}(\sigma_x) = 0,$$

$$\text{tr}(\sigma_z \sigma_x) = i \text{tr}(\sigma_y) = 0.$$

Thus under the Hilbert-Schmidt inner product, the Pauli matrices are mutually orthogonal. They are also orthogonal to the identity because $(I, \sigma_i) = \text{tr}(\sigma_i) = 0 \implies$ they form a basis for $L_{\mathbb{C}^2}$ since they are a set of 4 linearly independent vectors that span the space (since $\dim(L_{\mathbb{C}^2}) = 4$). To obtain an orthonormal basis, note that $\mathbb{1}^\dagger \mathbb{1} = \mathbb{1}$ and $\sigma_i^\dagger \sigma_i = \mathbb{1}$. Therefore

$$(I, I) = \text{tr}(I) = 2, \quad (\sigma_i, \sigma_i) = \text{tr}(I) = 2.$$

Each basis matrix has norm $\sqrt{2}$, so the complete orthonormal basis is

$$\left\{ \frac{I}{\sqrt{2}}, \frac{\sigma_x}{\sqrt{2}}, \frac{\sigma_y}{\sqrt{2}}, \frac{\sigma_z}{\sqrt{2}} \right\}.$$

1.4 Properties of the Trace Operation

Some facts about the trace operation

- **Lemma 1.1.** The trace operation is linear over complex scalars i.e. $\text{tr}(\alpha A + \beta B) = \alpha \text{tr}(A) + \beta \text{tr}(B)$
- **Lemma 1.2.** The trace operation of a product of linear operators is invariant under cyclic permutations i.e.

TODO: Add more lemmas about trace

2 Linear Operators and Observables

Definition 2.1. The *outer product* of two vectors $|u\rangle$ and $|v\rangle$ is the operator $|u\rangle\langle v|$. It acts on any vector $|w\rangle$ by

$$(|u\rangle\langle v|)|w\rangle = \langle v|w\rangle |u\rangle.$$

That is, it computes the overlap with $|v\rangle$, then multiplies $|u\rangle$ by that scalar. Note that the trace of the outer product is equal to the inner product i.e. $\text{tr}(|u\rangle\langle v|) = \langle u|v\rangle$. In the case where $|u\rangle = |v\rangle$ and $|u\rangle$ is normalized i.e. $\langle u|u\rangle = 1$, then $|u\rangle\langle u|$ is the projection operator onto the direction of $|u\rangle$

Definition 2.2. A *linear operator* $\hat{O}$ as a function that maps the vector space $\mathbb{V}$ to itself which is linear in its inputs:

$$\hat{O} \left(\sum_i a_i |v_i\rangle \right) = \sum_i a_i \hat{O}(|v_i\rangle)$$

Assuming $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ is complete orthonormal basis for V , then each linear operator has a matrix representation

$$\hat{O} = \sum_{i=1}^n \sum_{j=1}^n O_{ij} |i\rangle\langle j| \quad , \quad O_{ij} = \langle i|\hat{O}|j\rangle$$

In other words, a linear operator is defined by its action on a basis. If

$$|v\rangle = \sum_i^n v_i |i\rangle$$

then

$$\begin{aligned} \hat{O}|v\rangle &= \sum_{i=1}^n \sum_{j=1}^n (O_{ij} |i\rangle\langle j|) |v\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n O_{ij} (v_j |i\rangle) \end{aligned}$$

Not also that

$$\left(\sum_i |i\rangle\langle i| \right) |v\rangle = \sum_i v_i |i\rangle = \mathbb{1} \quad \implies \quad \sum_i |i\rangle\langle i| = \mathbb{1}$$

We can define the product of two operators $\hat{M}, \hat{N} \in \mathbb{L}_{\mathbb{V}}$ as

$$\hat{M}\hat{N} = \sum_{i,j,k} M_{ij} N_{jk} |i\rangle\langle k|.$$

Definition 2.3. The *commutator* of two linear operators $\hat{M}, \hat{N} \in \mathbb{L}_{\mathbb{V}}$ is

$$[\hat{M}, \hat{N}] = \hat{M}\hat{N} - \hat{N}\hat{M},$$

and the *anticommutator* is

$$\{\hat{M}, \hat{N}\} = \hat{M}\hat{N} + \hat{N}\hat{M}.$$

Definition 2.4. The *Hermitian conjugate*, or adjoint, of an operator is

$$\hat{O}^\dagger = \sum_{i,j} O_{ji}^* |i\rangle\langle j|.$$

In matrix notation,

$$O^\dagger = (O^T)^* = (O^*)^T, \quad (O^\dagger)_{ij} = O_{ji}^*.$$

This follows from conjugating the coefficient and reversing the outer product:

$$(O_{ij}|i\rangle\langle j|)^\dagger = O_{ij}^*|j\rangle\langle i|.$$

In other words, $\hat{O}^\dagger$ is the operator that satisfies $\langle u|\hat{O}v\rangle = \langle \hat{O}^\dagger u|v\rangle$ for all $|u\rangle$ and $|v\rangle$.

Some facts about Hermitian conjugates, trace, etc.

- **Lemma 2.1.** For two linear operators $\hat{A}$ and $\hat{B}$,

$$(AB)^\dagger = ((AB)^*)^T = (A^*B^*)^T = (B^*)^T(A^*)^T = B^\dagger A^\dagger.$$

Thus, if both $\hat{A}$ and $\hat{B}$ are Hermitian, then $(\hat{A}\hat{B})^\dagger = \hat{B}\hat{A} \implies$ their product is Hermitian if and only if they commute:

- **Lemma 2.2.** The Hermitian adjoint is linear on vectors and conjugate linear over complex scalars i.e. $\hat{A}^\dagger(\alpha|u\rangle + \beta|v\rangle) = \alpha\hat{A}^\dagger|u\rangle + \beta\hat{A}^\dagger|v\rangle$. and $(\alpha\hat{A} + \beta\hat{B})^\dagger = \alpha^*\hat{A}^\dagger + \beta^*\hat{B}^\dagger$.

TODO: Add more lemmas

Definition 2.5. An operator $\hat{O}$ is *Hermitian* if and only if $\hat{O} = \hat{O}^\dagger$. Every physical observable in quantum mechanics corresponds to a Hermitian operator.

Every Hermitian operator can be diagonalized with respect to an orthonormal eigenbasis $\{|i\rangle\}$

$$\hat{O} = \sum_i O_i |i\rangle\langle i|$$

This is called the *spectral decomposition* of $\hat{O}$, and the proof follows from the following theorems.

Theorem 2.1. Spectral Decomposition Theorem A linear operator $\hat{O}$ on V is diagonal with respect to some orthonormal basis of V if and only if $\hat{O}$ is normal, i.e. $\hat{O}^\dagger\hat{O} = \hat{O}\hat{O}^\dagger$

Theorem 2.2. A normal matrix is Hermitian if and only if it has real eigenvalues

Theorem 2.3. Eigenkets corresponding to different eigenvalues of a Hermitian matrix are orthogonal

Theorem 2.4. A positive linear operator must be Hermitian. Here positive means

$$\langle v|\hat{O}|v\rangle \geq 0$$

for every $|v\rangle \in V$. In particular, every such expectation value is real. Hermiticity is not assumed in this definition.

Theorem 2.5. Simultaneous diagonalization theorem Two Hermitian linear operators $\hat{A}$ and $\hat{B}$ can be diagonalized simultaneously if and only if

$$[\hat{A}, \hat{B}] = 0.$$

In other words, there is a complete orthonormal basis of simultaneous eigenkets:

$$\hat{A}|a, b\rangle = a|a, b\rangle, \quad \hat{B}|a, b\rangle = b|a, b\rangle. \quad (1.55)$$

3 Projective Measurements

Definition 3.1. A *projection operator* or *projector* is a linear operator $\hat{P} = \hat{P}^2$. For our purposes, we also require that $\hat{P} = \hat{P}^\dagger$ is Hermitian. In other words, the range and kernel of $\hat{P}$ are orthogonal.

Lemma 3.1. A Hermitian operator $\hat{M}$ can be diagonalized with respect to an orthonormal eigenbasis $\{|j\rangle\}$ such that

$$\hat{M} = \sum_j \lambda_j |j\rangle\langle j|.$$

If we use the set $\{m\}$ to label all different (real) eigenvalues of matrix $\hat{M}$, we can rewrite the above diagonal form as

$$\hat{M} = \sum_m m \hat{P}_m, \quad \hat{P}_m = \sum_j \delta_{\lambda_j, m} |j\rangle\langle j|.$$

In other words, $\hat{P}_m$ projects onto the subspace spanned by the eigenvectors that share the same eigenvalue.

Definition 3.2. The *expectation value* $\langle \hat{M} \rangle$ of an operator $\hat{M}$ with respect to a state $|\psi\rangle$ is

$$\langle \hat{M} \rangle \equiv \frac{\langle \psi | \hat{M} | \psi \rangle}{\langle \psi | \psi \rangle}$$

A consequence of this is the **Born Rule**, which states that for a physical observable represented by the Hermitian operator $\hat{M}$ and a state $|\psi\rangle$. Thus the probability of a measurement producing the eigenvalue λ_i is

$$\mathbf{P}(\lambda_i) = \langle \psi | \hat{P}_i | \psi \rangle = \langle \psi | \hat{P}_i^2 | \psi \rangle = \langle \hat{P}_i \psi | \hat{P}_i \psi \rangle = \|\hat{P}_i |\psi\rangle\|^2$$

Furthermore, the state collapses into a new state

$$\frac{\hat{P}_i |\psi\rangle}{\sqrt{\langle \psi | \hat{P}_i | \psi \rangle}}$$

after measurement.

4 Qubit and Bloch Sphere

4.1 Qubits

Definition 4.1. A *qubit* is a two state system associated with Hilbert space $\mathbb{C}^2$. Most commonly, we can think of it as the space of spin states for an electron. The conventions $|0\rangle$ and $|\uparrow\rangle$ are used to denote the spin up state i.e $s = \frac{\hbar}{2}$. $|1\rangle$ and $|\downarrow\rangle$ are used to denote the spin down state i.e. $s = -\frac{\hbar}{2}$ state.

Recall the 3 Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

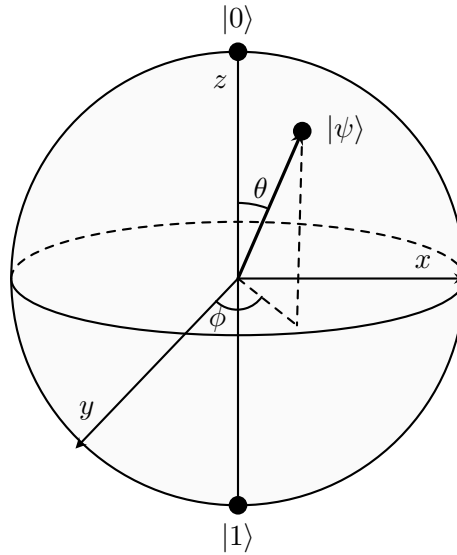
Their eigenvalues are ± 1 and their associated normalized eigenvectors are

$$\chi_{\pm}^{(x)} = \left\{ \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}, \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \right\}, \quad \chi_{\pm}^{(y)} = \left\{ \begin{pmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix}, \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix} \right\}, \quad \chi_{\pm}^{(z)} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

For any arbitrary normalized state $|\psi\rangle = a|0\rangle + b|1\rangle$, we can express it in terms of the eigenvectors of σ_x or σ_y :

$$|\psi\rangle = \left(\frac{a+b}{\sqrt{2}}\right) \chi_+^{(x)} + \left(\frac{a-b}{\sqrt{2}}\right) \chi_-^{(x)} = \left(\frac{a-ib}{\sqrt{2}}\right) \chi_+^{(y)} + \left(\frac{a+ib}{\sqrt{2}}\right) \chi_-^{(y)}$$

4.2 The Bloch Sphere



The Bloch sphere is useful because it turns a two-level quantum state into geometry: measurement probabilities become projections, unitary evolution becomes rotation, and mixedness becomes distance from the center.

4.2.1 Visualizing a state as a point on the sphere

Choose $|0\rangle = |\uparrow\rangle$ and $|1\rangle = |\downarrow\rangle$. Any normalized pure qubit state, after removing an overall phase, can be written as

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

Its Bloch vector is

$$\mathbf{r} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta).$$

which is the unit radial vector in spherical coordinates. Here θ is the polar angle measured from $+z$, and ϕ is the azimuthal angle.

The half-angle in the ket makes it so that

- (i) $|0\rangle$ and $|1\rangle$ are the north and south poles
- (ii) $(|0\rangle \pm |1\rangle)/\sqrt{2}$ point along $\pm x$ (often also notated $|\rightarrow\rangle$ and $|\leftarrow\rangle$ or $|+\rangle$ and $|-\rangle$)
- (iii) $(|0\rangle \pm i|1\rangle)/\sqrt{2}$ point along $\pm y$
- (iv) The Pauli expectations are equal to the cooresponding coordinate of the Bloch vector i.e. the projection onto the x, y, z axes

$$\langle \hat{X} \rangle = \sin \theta \cos \phi \quad , \quad \langle \hat{Y} \rangle = \sin \theta \sin \phi \quad , \quad \langle \hat{Z} \rangle = \cos \theta$$

Orthogonal kets correspond to antipodal points on the sphere, not perpendicular Bloch vectors. Multiplying a ket by an overall phase leaves its point unchanged while changing the relative phase moves it.

Other useful facts:

- (i) θ controls the measurement probabilities in the z -basis i.e.

$$P(Z = +1) = \cos^2(\theta/2) \quad , \quad P(Z = -1) = \sin^2(\theta/2)$$

5 Composite Quantum Systems

5.1 Tensors

Definition 5.1. Give a vector space V of dimension m and a vector space W of dimension n , their **tensor product space** is the vector space

$$V \otimes W \equiv \text{span}\{|v\rangle \otimes |w\rangle \mid |v\rangle \in V, |w\rangle \in W\}$$

of dimension mn . An element $|\psi\rangle$ of this vector space is called a **tensor** and takes the form

$$|\psi\rangle = \sum_i a_i (|v_i\rangle \otimes |w_i\rangle)$$

The tensor product is **bilinear** e.g.

$$|\psi\rangle = \sum_i a_i (|v_i\rangle \otimes |w_i\rangle) = \sum_i (a_i |v_i\rangle) \otimes |w_i\rangle = \sum_i |v_i\rangle \otimes (a_i |w_i\rangle).$$

The inner product on the tensor product space is defined

$$(\langle v| \otimes \langle w|) (|v'\rangle \otimes |w'\rangle) = \langle v|v'\rangle \langle w|w'\rangle$$

Since the inner product is conjugate-linear in the first argument and linear in the second, if

$$|\Psi\rangle = \sum_i a_i |v_i\rangle \otimes |w_i\rangle \quad , \quad |\Phi\rangle = \sum_j b_j |v'_j\rangle \otimes |w'_j\rangle$$

then

$$\langle \Psi | \Phi \rangle = \sum_{i,j} a_i^* b_j \langle v_i | v'_j \rangle \langle w_i | w'_j \rangle.$$

For linear operators $\hat{A}$ on V and $\hat{B}$ on W , we can construct a linear operator in the tensor product space

$$(\hat{A} \otimes \hat{B})(|v\rangle \otimes |w\rangle) = \hat{A}|v\rangle \otimes \hat{B}|w\rangle$$

TODO: Add explicit examples with matrices for clarity

5.2 Generalized Measurements

One way we can glean information about a system is to let it interact with another system that we can measure (called the ancilla). Consider two Hilbert spaces $\mathbb{V}_s$ (measured system) and $\mathbb{V}_a$ (ancilla system) and a state $|\Psi\rangle$ in their tensor product space

$$|\Psi\rangle \equiv |\psi\rangle_s \otimes |\phi\rangle_a$$

evolving unitarily under a Hamiltonian $\hat{H}$ for time t . Suppose we perform a projective measurement on $\mathbb{V}_a$ associated with the linear operator $\mathbb{1} \otimes \hat{P}_a$, where $\hat{P}_a$ has the spectral decomposition

$$\hat{P}_a = \sum_q \lambda_q |q\rangle \langle q|$$

and $\{|q\rangle\}$ forms a complete orthonormal basis for $\mathbb{V}_a$. Immediately before the measurement, the joint state is

$$|\Psi'\rangle = \hat{U}(|\psi\rangle_s \otimes |\phi\rangle_a)$$

where $\hat{U} = e^{-i\hat{H}t/\hbar}$. After such a measurement with measurement outcome λ_q , the state collapses into a new (unnormalized) state $\hat{M}_q(|\psi\rangle_s \otimes |\phi\rangle_a)$ where

$$\hat{M}_q \equiv {}_a\langle q | \hat{U} | \phi \rangle_a \in \mathbb{L}_{\mathbb{V}_s}$$

is a linear operator on $\mathbb{V}_s$. In other words, $\hat{M}_q$ describes the effect on the first system associated with obtaining a measurement outcome λ_q from the ancilla system. More explicitly,

$$\hat{M}_q |\psi\rangle_s = (\mathbb{1}_s \otimes {}_a\langle q |) \hat{U} (|\psi\rangle_s \otimes |\phi\rangle_a)$$

Reading from right to left, this means that we can obtain $\hat{M}_q |\psi\rangle_s$ by

1. Applying the joint evolution operator $\hat{U}$ to the initial joint state

$$\hat{U}(|\psi\rangle_s \otimes |\phi\rangle_a) = \sum_{i,j} c_{ij} (|i\rangle_s \otimes |j\rangle_a)$$

2. Taking the inner product

$$\left\langle (\mathbb{1}_s \otimes {}_a\langle q|) \left| \sum_{i,j} c_{ij} (|i\rangle_s \otimes |j\rangle_a) \right. \right\rangle = \sum_{i,j} c_{ij} |i\rangle_s \underbrace{{}_a\langle q|j\rangle_a}_{\delta_{qj}} = \sum_{i,j} c_{ij} \delta_{qj} |i\rangle_s = \sum_i c_{iq} |i\rangle_s$$

Since

$$\begin{aligned} \hat{P}_q |\Psi'\rangle &= \sum_j (\hat{M}_j |\psi\rangle_s) \otimes (|q\rangle_a {}_a\langle q|j\rangle_a) \\ &= \sum_j \delta_{qj} (\hat{M}_j |\psi\rangle_s) \otimes |q\rangle_a \\ &= \hat{M}_q (|\psi\rangle_s \otimes |q\rangle_a) \end{aligned}$$

by the Born rule, the probability that a measurement on $\mathbb{V}_a$ produces q is

$$\mathbf{P}(q) = \langle \Psi' | \hat{P}_q | \Psi' \rangle = {}_s\langle \psi | \hat{M}_q^\dagger \hat{M}_q | \psi \rangle_s$$

thus the $\hat{M}_q$'s satisfy the completeness relation $\sum_q \hat{M}_q^\dagger \hat{M}_q = \mathbb{1}_s$. Alternatively, we can also write

$$\mathbf{P}(q) = \text{tr}(\hat{M}_q |\psi\rangle_s \langle \psi|_s \hat{M}_q^\dagger)$$

Example 5.1. TODO: Add examples for clarity

6 Density Matrices, Mixed States, Entanglement

Definition 6.1. Suppose we have a system that is described by normalized (but not necessarily orthogonal) states $|\psi_i\rangle$ with probability p_i . This system can then be described by a **density matrix**

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle \psi_i|$$

Then, the expectation value of any operator $\hat{A}$ is

$$\langle \hat{A} \rangle = \sum_i p_i \langle \psi_i | \hat{A} | \psi_i \rangle = \text{tr}(\hat{\rho} \hat{A})$$

If the system is described by a specific state ψ , we call it a **pure state** $\iff$

$$\hat{\rho} = |\psi\rangle \langle \psi| \quad , \quad \hat{\rho}^2 = \hat{\rho}$$

is simply the projection operator. A state that is not pure is called a **mixed state**.

6.1 Properties of the Density Matrix

The density matrix has the following properties:

- (i) It is self-adjoint: $\hat{\rho} = \hat{\rho}^\dagger$.

(ii) It has unit trace: $\text{tr}(\hat{\rho}) = 1$.

$$\begin{aligned}\text{tr}(\hat{\rho}) &= \sum_n \langle n | \hat{\rho} | n \rangle = \sum_n \sum_i p_i \langle n | \psi_i \rangle \langle \psi_i | n \rangle \\ &= \sum_i p_i \sum_n \langle \psi_i | n \rangle \langle n | \psi_i \rangle = \sum_i p_i \langle \psi_i | \psi_i \rangle = 1.\end{aligned}$$

Here, in an intermediate step, we needed to use the decomposition of unity $\sum_n |n\rangle \langle n| = \mathbb{1}$ and, in the final step, we used the fact that our states $|\psi_i\rangle$ are normalised. We see that the fact the density matrix has unit trace is equivalent to the normalisation of a probability distribution, $\sum_i p_i = 1$.

(iii) It is non-negative i.e. $\hat{\rho} \geq 0$: $\langle \phi | \hat{\rho} | \phi \rangle \geq 0$ for all $|\phi\rangle \in \mathcal{H}$. This is equivalent to the requirement that $p_i \geq 0$.

Any operator $\hat{\rho}$ which satisfies these three properties can be viewed as a density matrix for a quantum system. To see this, suppose $\{|\phi_n\rangle\}$ are eigenvectors of $\hat{\rho}$ with eigenvalues $\{p_n\}$. Then

$$\hat{\rho} |\phi_n\rangle = p_n |\phi_n\rangle$$

Because $\hat{\rho} = \hat{\rho}^\dagger$, we know that $p_n \in \mathbb{R}$. Properties (ii) and (iii) above $\implies \sum_n p_n = 1$ and $p_n \geq 0 \implies$ we can interpret $\{p_n\}$ as a probability distribution and write $\hat{\rho}$ as

$$\hat{\rho} = \sum_n p_n |\phi_n\rangle \langle \phi_n|$$

TODO: Add how it relates back to the Bloch sphere (pages 1067-1070 in Tong)

6.2 Entanglement

Definition 6.2. A Hilbert space $\mathcal{H}$ has a **bipartite decomposition** if it can be expressed as the tensor product space of two Hilbert spaces $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$.

If $\{|i\rangle_A\}$ and $\{|j\rangle_B\}$ are orthonormal bases for $\mathcal{H}_A$ and $\mathcal{H}_B$, then a general composite state $|\Psi\rangle \in \mathcal{H}$ has the form

$$|\Psi\rangle = \sum_{i,j} c_{ij} (|i\rangle_A \otimes |j\rangle_B)$$

All measurements on only $\mathcal{H}_A$ correspond to operators of the form $\hat{O} = \hat{A} \otimes \mathbb{1}$. If $\hat{\rho}_{AB}$ is the density matrix for the full system, then the expectation value of

$$\langle \hat{A} \rangle = \text{tr}_A \text{tr}_B \left((\hat{A} \otimes \mathbb{1}) \hat{\rho}_{AB} \right) = \text{tr}_A (\hat{A} \hat{\rho}_A)$$

where $\hat{\rho}_A = \text{tr}_B(\hat{\rho}_{AB})$ is the **reduced density matrix** which is obtained by taking the partial trace over $\mathcal{H}_B$.

Example 6.1. Consider two matrices $A^{m \times m}$ and $B^{n \times n}$ (which could represent linear operators).
Then

$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1m}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mm}B \end{pmatrix}$$

and $\text{tr}(A \otimes B) = a_{11} \text{tr}(B) + \cdots + a_{mm} \text{tr}(B) = \text{tr}(A) \text{tr}(B)$ is a scalar, but the partial traces

$$\text{tr}_B(A \otimes B) = \begin{pmatrix} a_{11} \text{tr}(B) & \cdots & a_{1m} \text{tr}(B) \\ \vdots & \ddots & \vdots \\ a_{m1} \text{tr}(B) & \cdots & a_{mm} \text{tr}(B) \end{pmatrix} = \text{tr}(B)A$$

$$\text{tr}_A(A \otimes B) = \begin{pmatrix} b_{11} \text{tr}(A) & \cdots & b_{1n} \text{tr}(A) \\ \vdots & \ddots & \vdots \\ b_{n1} \text{tr}(A) & \cdots & b_{nn} \text{tr}(A) \end{pmatrix} = \text{tr}(A)B$$

are matrices.

TODO: Add Schmidt decomposition